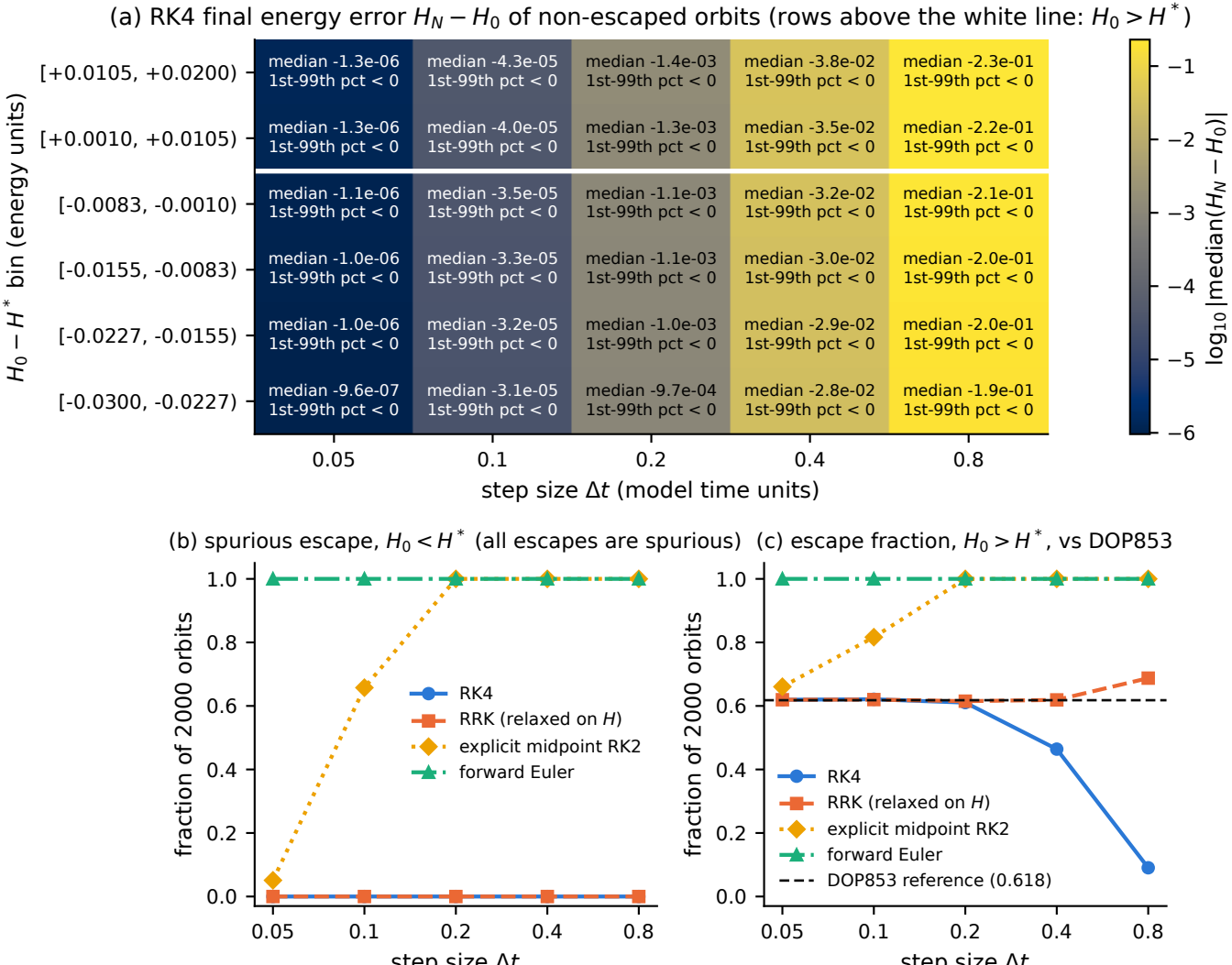
(a) RK4 final energy error $H_N - H_0$ of non-escaped orbits (rows above the white line: $H_0 > H^*$)
$H_0 - H^*$ bin (energy units)
[+0.0105, +0.0200)
[+0.0010, +0.0105)
[-0.0083, -0.0010)
[-0.0155, -0.0083)
[-0.0227, -0.0155)
[-0.0300, -0.0227)
median -1.3e-06 1st-99th pct < 0
median -4.3e-05 1st-99th pct < 0
median -1.4e-03 1st-99th pct < 0
median -3.8e-02 1st-99th pct < 0
median -2.3e-01 1st-99th pct < 0
median -1.3e-06 1st-99th pct < 0
median -4.0e-05 1st-99th pct < 0
median -1.3e-03 1st-99th pct < 0
median -3.5e-02 1st-99th pct < 0
median -2.2e-01 1st-99th pct < 0
median -1.1e-06 1st-99th pct < 0
median -3.5e-05 1st-99th pct < 0
median -1.1e-03 1st-99th pct < 0
median -3.2e-02 1st-99th pct < 0
median -2.1e-01 1st-99th pct < 0
median -1.0e-06 1st-99th pct < 0
median -3.3e-05 1st-99th pct < 0
median -1.1e-03 1st-99th pct < 0
median -3.0e-02 1st-99th pct < 0
median -2.0e-01 1st-99th pct < 0
median -1.0e-06 1st-99th pct < 0
median -3.2e-05 1st-99th pct < 0
median -1.0e-03 1st-99th pct < 0
median -2.9e-02 1st-99th pct < 0
median -2.0e-01 1st-99th pct < 0
median -9.6e-07 1st-99th pct < 0
median -3.1e-05 1st-99th pct < 0
median -9.7e-04 1st-99th pct < 0
median -2.8e-02 1st-99th pct < 0
median -1.9e-01 1st-99th pct < 0
0.05
0.1
0.2
0.4
0.8
step size Δt (model time units)
$\log_{10}|\text{median}(H_N - H_0)|$
−1
−2
−3
−4
−5
−6
(b) spurious escape, $H_0 < H^*$ (all escapes are spurious)
(c) escape fraction, $H_0 > H^*$, vs DOP853
fraction of 2000 orbits
1.0
0.8
0.6
0.4
0.2
0.0
RK4
RRK (relaxed on H)
explicit midpoint RK2
forward Euler
DOP853 reference (0.618)
step size Δt